Multiple Sequence Alignment

Introduction

Multiple Sequence Alignment (MSA) is a bioinformatics technique used to compare protein sequences from different organisms. It helps identify conserved regions, evolutionary relationships, and functional similarities. In this study, insulin protein sequences from Human, Pig, Rat, Mouse, and Chicken were aligned using Clustal Omega.

Objective

To perform Multiple Sequence Alignment of insulin protein sequences from different organisms and identify conserved amino acid regions.

Materials and Software Used

- UniProt Database
- Clustal Omega
- Internet Browser

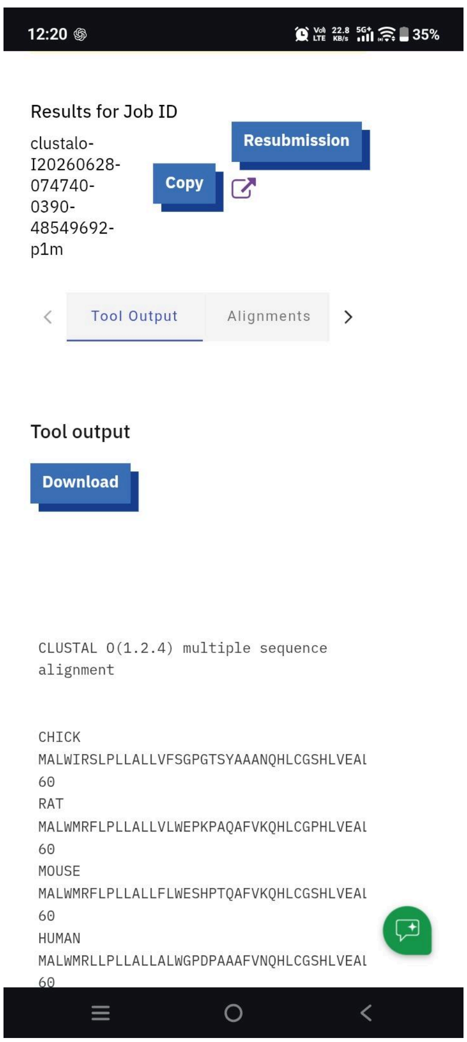
Methodology

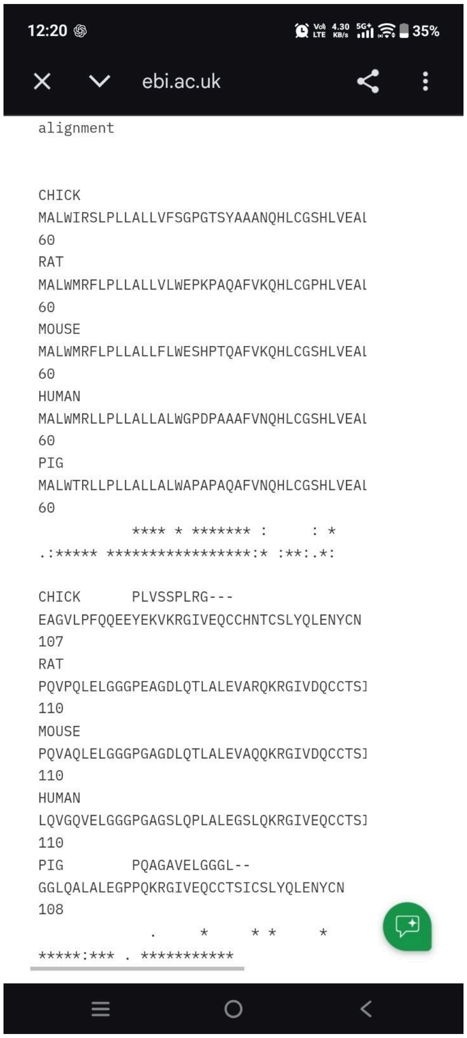
1.Retrieved insulin protein sequences from UniProt.
2.Copied FASTA sequences of Human, Pig, Rat, Mouse, and Chicken.
3.Submitted the sequences to Clustal Omega.
4.Performed Multiple Sequence Alignment.
5.Analyzed conserved regions and sequence similarities.

Results and Discussion

The alignment showed that insulin protein sequences from Human, Pig, Rat, Mouse, and Chicken share several conserved amino acid residues. Human, Rat, Mouse, and Pig exhibited high sequence similarity, while Chicken showed a few differences, reflecting evolutionary divergence. The conserved regions indicate the functional importance of the insulin protein.

Screenshot 1
Multiple Sequence Alignment Result



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alignment


CHICK
MALWIRSLPLLALLVFSGPGTSYAAANQHLCGSHLVEAL
60
RAT
MALWMRFLPLLALLVLWEPKPAQAFVKQHLCGPHLVEAL
60
MOUSE
MALWMRFLPLLALLFLWESHPTQAFVKQHLCGSHLVEAL
60
HUMAN
MALWMRLLPLLALLALWGPDPAAAFVNQHLCGSHLVEAL
60
PIG
MALWTRLLPLLALLALWAPAPAQAFVNQHLCGSHLVEAL
60
           **** * ******* :     : *
.:***** *****************:* :**:.*:

CHICK      PLVSSPLRG---
EAGVLPFQQEEYEKVKRGIVEQCCHNTCSLYQLENYCN
107
RAT
PQVPQLELGGGPEAGDLQTLALEVARQKRGIVDQCCTSI
110
MOUSE
PQVAQLELGGGPGAGDLQTLALEVAQQKRGIVDQCCTSI
110
HUMAN
LQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSI
110
PIG        PQAGAVELGGGL--
GGLQALALEGPPQKRGIVEQCCTSICSLYQLENYCN
108
             .     *     * *     *
*****:*** . ***********
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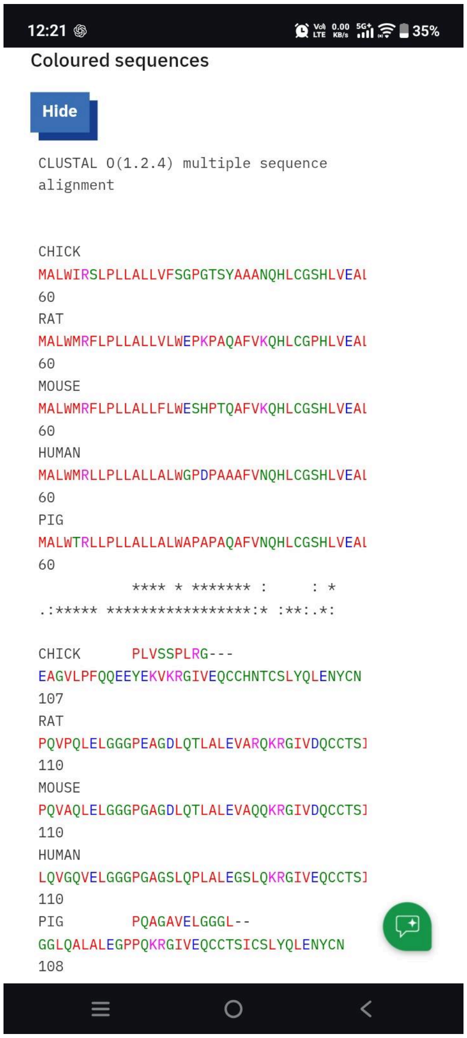
Alignment of insulin protein sequences from Human, Pig, Rat, Mouse, and Chicken showing conserved amino acid residues and sequence similarities.

Screenshot 2

Colour-Coded Sequence Alignment



12:21 35%

Coloured sequences

Hide

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CLUSTAL O(1.2.4) multiple sequence
alignment


CHICK
MALWIRSLPLLALLVFSGPGTSYAAANQHLCGSHLVEAL
60
RAT
MALWMRFLPLLALLVLWEPKPAQAFVKQHLCGPHLVEAL
60
MOUSE
MALWMRFLPLLALLFLWESHPTQAFVKQHLCGSHLVEAL
60
HUMAN
MALWMRLLPLLALLALWGPDPAAAFVNQHLCGSHLVEAL
60
PIG
MALWTRLLPLLALLALWAPAPAQAFVNQHLCGSHLVEAL
60
           **** * ******* :     : *
.:***** *****************:* :**:.*:

CHICK      PLVSSPLRG---
EAGVLPFQQEEYEKVKRGIVEQCCHNTCSLYQLENYCN
107
RAT
PQVPQLELGGGPEAGDLQTLALEVARQKRGIVDQCCTSI
110
MOUSE
PQVAQLELGGGPGAGDLQTLALEVAQQKRGIVDQCCTSI
110
HUMAN
LQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSI
110
PIG        PQAGAVELGGGL--
GGLQALALEGPPQKRGIVEQCCTSICSLYQLENYCN
108
```

Colour-coded Multiple Sequence Alignment highlighting conserved regions among insulin protein sequences

Conclusion

Multiple Sequence Alignment successfully demonstrated that insulin proteins are highly conserved across different species. Clustal Omega effectively identified conserved regions and sequence similarities, making it a valuable tool for comparative protein analysis.

References

1.https://www.uniprot.org

2.https://www.ebi.ac.uk/Tools/msa/clustalo